

Notes about the post-XSS world

Notes about the post-XSS world - Michal Zalewski, Michal Zalewski.

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Content

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[Content Security Policy](#) is gaining steam, and we've seen a flurry of other complementary approaches that share a common goal: to minimize the impact of markup injection vulnerabilities by preventing the attacker from executing unauthorized JavaScript. We are so accustomed to thinking about markup injection in terms of [cross-site scripting](#) that we don't question this approach - but perhaps we should?

[This collection of notes](#) is a very crude thought experiment in imagining the attack opportunities in a post-XSS world. The startling realization I had by the end of that half-baked effort is that the landscape would not change that much: The hypothetical universal deployment of CSP places some additional constraints on what you can do, but the differences are not as substantial as you may suspect. In that sense, the frameworks are conceptually similar to DEP, stack canaries, or ASLR: They make your life harder, but reliably prevent exploitation far less frequently than we would have thought.

Credit where credit is due: The idea for writing down some of the possible attack scenarios comes from Mario Heiderich and Elie Bursztein, who are aiming to write a more coherent and nuanced academic paper on this topic, complete with vectors of their design, and some very interesting 0-day bugs; I hope to be able to contribute to that work. In the meantime, though, it seems that everybody else is thinking out loud about the same problems - including Devdatta Akhawe and Collin Jackson - so I thought that sharing the current notes may be useful, even if the observations are not particularly groundbreaking.